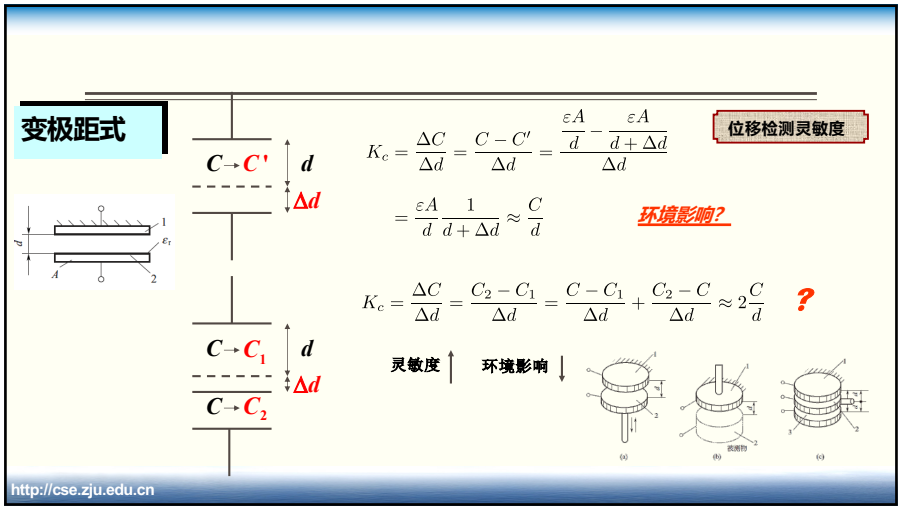
2.4 电容式检测元件

1



2



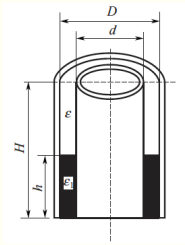
3

4

变介电常数式

$$K_c = \frac{\Delta C}{\Delta \varepsilon} = \frac{\frac{A \cdot \Delta \varepsilon}{d}}{\Delta \varepsilon} = \frac{A}{d}$$

☺ 可用来测水位、物位、介质厚度等



$$C = \frac{2\pi\varepsilon_1 h}{\ln(D/d)} + \frac{2\pi\varepsilon(H-h)}{\ln(D/d)} = \frac{2\pi\varepsilon H}{\ln(D/d)} + \frac{2\pi h(\varepsilon_1 - \varepsilon)}{\ln(D/d)}$$
$$= C_0 + \frac{2\pi h(\varepsilon_1 - \varepsilon)}{\ln(D/d)}$$

?

5

电容检测元件的等效电路

$$C_e = \frac{C}{1 - \omega^2 LC}$$

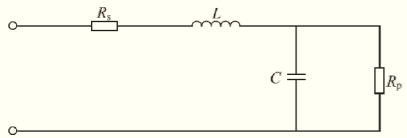


图 2.29 电容检测元件等效电路

$$\frac{\Delta C_e}{C_e} = \frac{\Delta C}{C}\left(\frac{1}{1 - \omega^2 LC}\right)$$

因此测量时必须与校准时处在同样条件下,其电源频率与引线长度不能改变。

6

消除温度对电容的影响

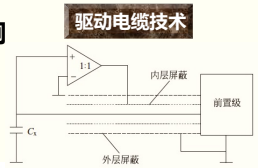
温度补偿

温度变化
- 几何尺寸变化 → 附加误差
  - 选用温度系数小且稳定的材料
- 介电常数变化 → 测量误差
  - 在转换电路中采取补偿措施

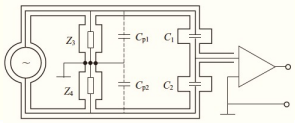
消除寄生电容的影响

- 增加初始电容
- 集成法
- 驱动电缆技术
- 整体屏蔽技术

驱动电缆技术

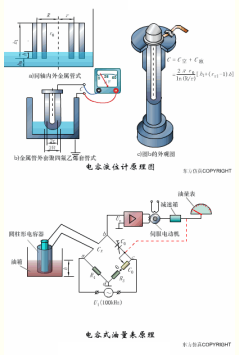


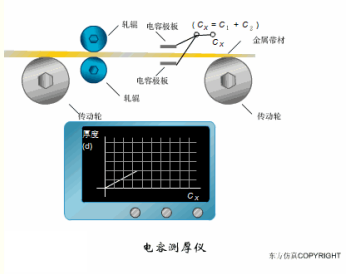
整体屏蔽技术



7

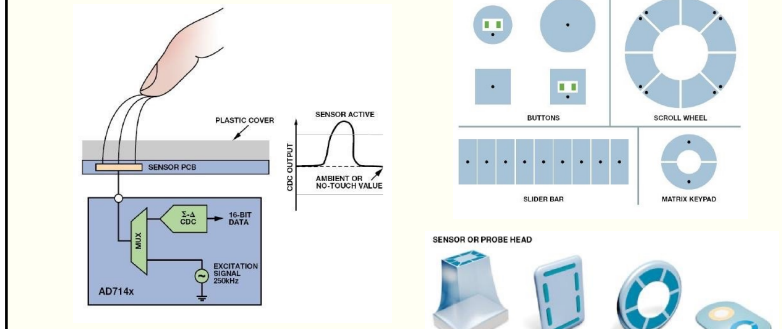
Applications





电容测厚仪

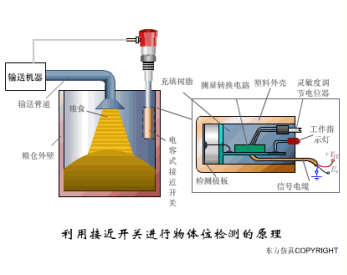
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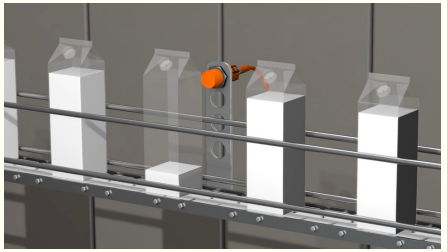




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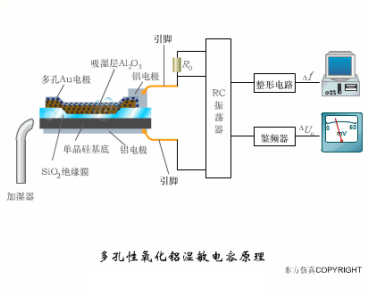
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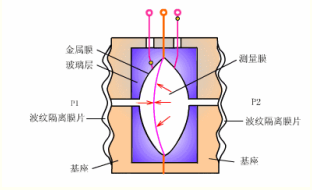




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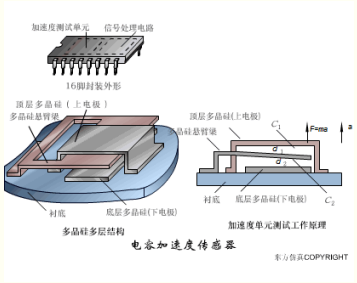
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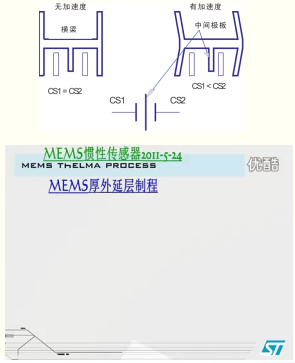




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11





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12

- **电容式检测元件的特点**
  - 结构简单
  - 低功耗
  - 动态特性好，兆赫兹
  - 非接触测量

  - 电容小，电容变化量小
  - 寄生/杂散电容干扰
  - 外界环境干扰

讨论题：

- **变极距型电容传感器的非线性及其补偿。**
- **如何实现电容值的测量？**